

Literature status: the 1-Schur versus 1-strong Schur questions

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Status. Both focused questions at the end of arXiv:1103.2975 are answered by arXiv:2505.12893. This note records the exact correspondence and does not claim a new mathematical result.

Definitions. For a bounded sequence (x_n) in a Banach space X , write

$$\text{ca}(x_n) = \inf_N \text{diam}\{x_n : n \geq N\}, \quad \delta(x_n) = \text{diam clust}_{w^*}(x_n) \subset X^{**}.$$

The space has the 1-Schur property when $\text{ca}(x_n) \leq \delta(x_n)$ for every bounded sequence. It has the 1-strong Schur property when every normalized δ -separated sequence has, for every $\varepsilon > 0$, a subsequence $(2/\delta + \varepsilon)$ -equivalent to the standard basis of ℓ_1 .

The two source questions. In its final remarks, arXiv:1103.2975 proves for real Banach spaces that

$$1\text{-Schur} \implies 1\text{-strong Schur} \implies 5\text{-Schur}.$$

It asks whether the first two properties are equivalent and conjectures that they are not. It separately notes that the proof of the first implication is real and asks, in effect, whether it remains valid over the complex field.

Exact later answer in the real case. Section 8 of O. F. K. Kalenda, *Quantitative Schur property and measures of weak non-compactness*, arXiv:2505.12893, gives the following example.

Real counterexample. Let $M = \mathbb{Z}$ with the shortest-path metric for the graph in which distinct m, n are adjacent exactly when $m \neq -n$. Thus

$$d(m, n) = \begin{cases} 0, & m = n, \\ 2, & m = -n \neq 0, \\ 1, & \text{otherwise.} \end{cases}$$

Then the real Lipschitz-free space $\mathcal{F}(M)$ has the 2-Schur property, fails the c -Schur property for every $c < 2$, and has the 1-strong Schur property.

In particular $\mathcal{F}(M)$ is 1-strong Schur but not 1-Schur. This is exactly the counterexample conjectured in arXiv:1103.2975.

Exact later answer in the complex case. The same paper also distinguishes the two notions in the opposite direction over the complex field.

Complex counterexample. *Let X be the canonical complexification of the real Banach space ℓ_1 (with the complexification norm used in the paper). Then X has the 1-Schur property, but it fails the quantitative strong-Schur condition with constant c for every $c < \pi/2$. In particular, it fails the 1-strong Schur property.*

The witness is the canonical-vector sequence. Its complex lower ℓ_1 constant is $2/\pi$, so no subsequence can have the constant required for $c < \pi/2$. Therefore the real implication 1-Schur $\Rightarrow$ 1-strong Schur is false for complex Banach spaces.

Verification and conclusion. The later paper cites arXiv:1103.2975 for the original definitions and implications, announces in its introduction that the first graph example “witnesses that these two properties are different also within real Banach spaces,” and proves the two examples above in full. The source’s conjecture and its complex-field uncertainty are therefore both resolved by existing literature. The target is classified as `literature_already_answered`.

References.

1. O. F. K. Kalenda and J. Spurny, *On a difference between quantitative weak sequential completeness and the quantitative Schur property*, Proc. Amer. Math. Soc. 140 (2012), 3435–3444; arXiv:1103.2975.
2. O. F. K. Kalenda, *Quantitative Schur property and measures of weak non-compactness*, Results Math. 80 (2025), article 233; arXiv:2505.12893.